Project Report
Healthcare Claims Cost & Utilization Analysis

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Tools Used: Excel, SQL, Tableau

Summary:

This analysis examines healthcare insurance claims data to identify the primary drivers of medical spending across claim types, procedures (CPT), diagnoses (ICD), providers, and members.

The analysis reveals that inpatient services are the dominant cost driver, accounting for the highest total paid amounts and highest average cost per claim. A small subset of members contributes a disproportionate share of total healthcare spending, largely driven by inpatient admissions.

Additionally, chronic conditions such as hypertension and diabetes drive significant long-term costs, while paid amounts are consistently lower than billed amounts, particularly for inpatient care, reflecting negotiated rates and claim adjustments.

These findings highlight opportunities for cost containment through care management, preventive programs, and provider contract optimization.

Dataset Overview:

The dataset contains synthetic healthcare claims data with the following fields:

- Member ID
- Claim Type (Inpatient, Emergency, Outpatient, Lab, Pharmacy)
- CPT Codes (procedures performed)
- ICD Codes (diagnoses)
- Provider ID
- Billed Amount
- Paid Amount
- Service Date

The dataset allows analysis across **services, procedures, diagnoses, providers, and members**.

Total Claims = 449

claim_id	member_id	provider_id	claim_date	claim_type	cpt_code	icd_code	billed_amount	paid_amount
1	1	PRV00001	12-03-2023	Outpatient	12345	A12.3	1800	1600
2	1	PRV00002	22-07-2023	Inpatient	67890	B99.4	32000	28000
3	1	PRV00003	15-01-2024	Lab	23456	C50.1	450	400
4	1	PRV00004	05-09-2023	Pharmacy	34567	D12.7	120	120
5	2	PRV00123	12-04-2023	Outpatient	12345	A12.3	2100	1680
6	2	PRV00456	20-06-2023	Inpatient	67890	B99.4	15250	0
7	2	PRV00789	02-11-2023	Emergency	54321	C44.9	5200	3900
8	2	PRV01234	14-02-2024	Pharmacy	11111	E11.40	320	320
9	3	PRV12345	15-03-2023	Outpatient	123	A12.3	2400	1800
10	3	PRV67890	20-04-2023	Inpatient	567	C45.01	12500	10250
11	3	PRV54321	05-06-2023	Emergency	345	J11.1	5200	4100
12	3	PRV98765	12-11-2023	Lab	999	N18.9	420	320
13	3	PRV24680	08-02-2024	Pharmacy	12	Z79.898	210	180
14	4	PRV00001	15-03-2023	Inpatient	10001	A00.0	42000	4000
15	4	PRV00002	22-06-2023	Outpatient	10023	B20.1	3200	2800
16	4	PRV00003	08-11-2023	Emergency	10112	C34.91	5600	4800
17	4	PRV00004	14-02-2024	Lab	10234	E11.9	250	200
18	4	PRV00005	30-07-2024	Pharmacy	10345	N18.6	180	180
19	5	PRV00001	15-03-2023	Outpatient	12345	A12.3	1200	980

Data Preparation and Validation:

1. Excel – Data Cleaning

Excel was used for initial data preparation:

- Standardized service date formats
- Removed duplicate claim records
- Checked and handled null values in key fields
- Verified numeric consistency for billed and paid amounts

This ensured data quality prior to analysis.

2. **SQL – Validation & Calculation Checks**

SQL was used to:

- Validate total billed and paid amounts
- Confirm claim counts by claim type
- Verify average paid amounts and aggregations

This step ensured that all Tableau visualizations were based on **accurate and verified calculations**.

Tableau Analysis and Calculated Fields:

The cleaned dataset was imported into Tableau for visualization and insight generation.

Calculated Fields Created:

- **Paid Ratio (%)**

  $$\text{Paid Ratio} = \frac{\text{SUM(Paid Amount)}}{\text{SUM(Billed Amount)}} \times 100$$

- **Average Paid Per Claim**

  $$\text{Average Paid Per Claim} = \frac{\text{Sum(Paid Amount)}}{\text{Number of Counts}}$$

Threshold Filtering:

To avoid distortion from rare one-time cases:

- CPT codes and providers were included only if they had 5 or more claims.
- This improves analytical reliability and interpretability.

Analysis and Findings:

1. Which claim types are the most expensive?

Claim Type	Total Billed	Total Paid	Avg Paid per Claim	Paid Ratio
Inpatient	1478601	1092456	11035	73.88%
Emergency	384242	294441	3346	76.63%
Outpatient	160718	129053	1229	80.3%
Lab	25790	23412	308	90.78%
Pharmacy	12815	11382	141	88.82%

Insight:
Inpatient claims represent the largest financial burden, driven by high cost per claim rather than volume. Despite having the lowest paid ratio, inpatient services dominate total healthcare spending.

2. Which CPT codes drives the highest spending?

Top contributors to overall spending:

- CPT 67890 → $242,735 (25 claims)
- CPT 23456 → $203,791 (29 claims)
- CPT 123 → $122,010 (12 claims)
- CPT 12345 → $115,237 (47 claims)
- CPT 99223 → $57,350 (6 claims)

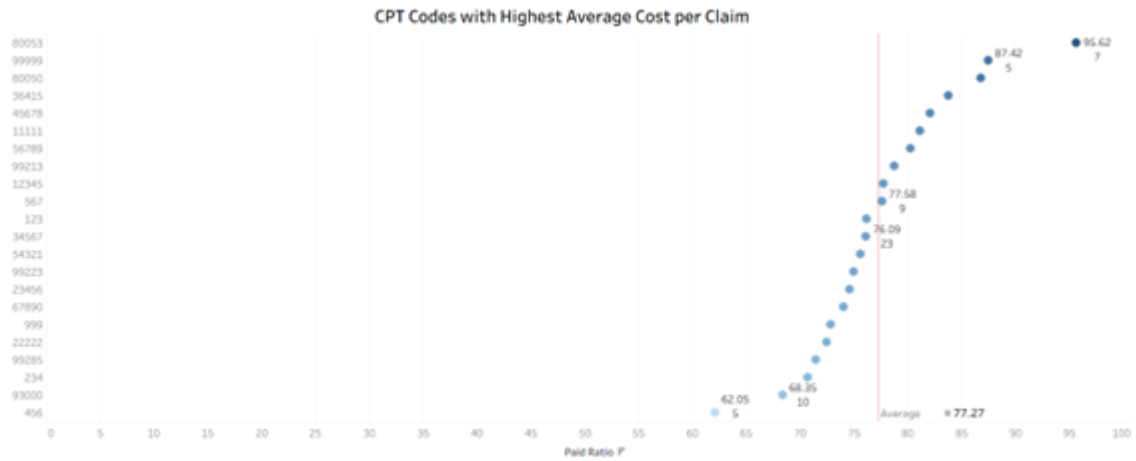
Insight: These CPTs drive cost mainly because they combine meaningful volume + high payments.

CPT Codes with Highest Average Cost per Claim (≥ 5 claims):

Reference line on chart: Average CPT paid ratio (≥5 claims) = 77.27%

CPT	# Claims	Avg Paid/Claim	Paid Ratio
123	12	10,167.50	76.12%
67890	25	9,709.40	74.00%
99223	6	9,558.33	74.97%
23456	29	7,027.27	74.54%
567	9	4,329.89	77.58%
54321	11	3,327.27	75.57%

Insight: These CPTs are "high cost" because each occurrence is expensive, not because their paid ratios are unusually high. In fact, the top three (123, 67890, 99223) sit around/below the average paid ratio line (77.27%), meaning the risk is cost intensity, not reimbursement generosity.



<u>Top ICD codes by Total Paid Amount</u>:

ICD I10(Hypertension) → $259,566
ICD A12.3(Pulmonary tuberculosis) → $152,147
ICD B20(Human immunodeficiency virus [HIV] disease) → $140,990
ICD B20.1(HIV disease resulting in mycobacterial infection) → $105,210
ICD C34.91(Malignant neoplasm of unspecified part of right bronchus or lung) → $62,905

Insight: Chronic and complex medical conditions are strongly associated with **inpatient admissions and long-term cost accumulation**.

3. Which members account for the largest share of total costs?

Member	Total Paid
6	43,300.00
32	40,080.00
58	35,920.00
82	30,690.00
28	30,560.00
20	30,375.00
71	30,230.00
1	30,120.00
36	25,530.75
8	23,046.80

Drivers of Healthcare Spending for High-Cost Members

Member Id	Claim T..	Paid Amount	% of Total Paid Amount along Claim Type
6	Inpatient	31,500	72.75%
	Emergency	8,000	18.48%
	Outpatient	2,600	6.00%
	Lab	1,200	2.77%
32	Inpatient	35,000	87.33%
	Emergency	3,200	7.98%
	Outpatient	1,500	3.74%
	Lab	320	0.80%
	Pharmacy	60	0.15%
58	Inpatient	25,000	69.60%
	Emergency	6,800	18.93%
	Outpatient	3,800	10.58%
	Pharmacy	320	0.89%
82	Inpatient	25,600	83.41%
	Emergency	3,600	11.73%
	Outpatient	1,280	4.17%
	Pharmacy	210	0.68%
28	Inpatient	25,600	83.77%
	Emergency	960	3.14%
	Outpatient	3,600	11.78%
	Lab	280	0.92%
	Pharmacy	120	0.39%
20	Inpatient	25,050	82.47%
	Emergency	3,000	9.88%
	Outpatient	1,710	5.63%
	Lab	465	1.53%
	Pharmacy	150	0.49%
71	Inpatient	25,000	82.70%
	Emergency	1,100	3.64%
	Outpatient	3,600	11.91%
	Lab	350	1.16%
	Pharmacy	180	0.60%
1	Inpatient	28,000	92.96%
	Outpatient	1,600	5.31%
	Lab	400	1.33%
	Pharmacy	120	0.40%
36	Inpatient	15,000	58.75%
	Emergency	7,500	29.38%
	Outpatient	2,500	9.79%
	Lab	320	1.25%
	Pharmacy	211	0.83%
8	Inpatient	17,251	74.85%
	Emergency	4,100	17.79%
	Outpatient	1,341	5.82%
	Lab	280	1.21%
	Pharmacy	76	0.33%

Breaks down the cost drivers for high-cost members by claim type and percentage contribution.

Drivers for high-cost members

- **Member 6:** Inpatient 31,500 (**72.75%**), Emergency 8,000 (**18.48%**), Outpatient 2,600 (**6.00%**), Lab 1,200 (**2.77%**)
- **Member 1:** Inpatient 28,000 (**92.96%**) dominates total cost

Conclusion: A small subset of members drives a disproportionate share of spend, and for most of them the main driver is **Inpatient**.

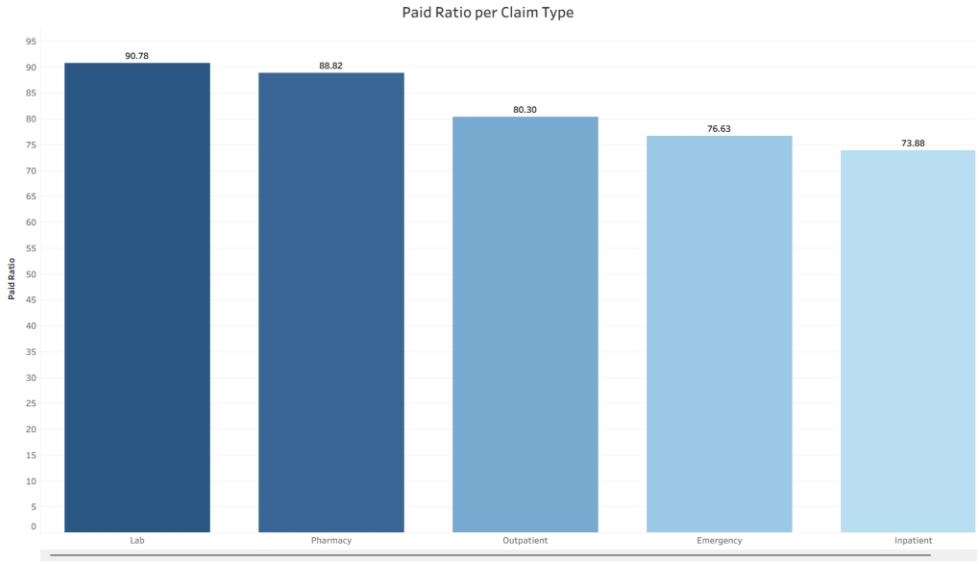
4. How do billed amounts compare to paid amounts?

By Claim Type:

Paid ratios show how much of billed is reimbursed:

- **Highest paid ratios:** Lab (**90.78%**), Pharmacy (**88.82%**)
- **Lowest paid ratio:** Inpatient (**73.88%**)

Meaning: Inpatient has the largest "gap" between billed and paid, suggesting heavier negotiation/adjustments/denials.



<u>By provider (≥ 5 claims</u>):

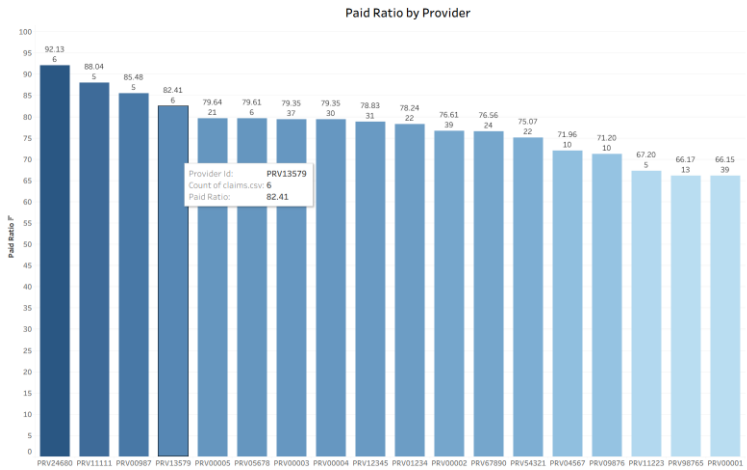
Highest paid ratios (examples):

- PRV24680: 92.13% (6 claims)
- PRV11111: 88.04% (5 claims)
- PRV00987: 85.48% (5 claims)

Lowest paid ratios (examples):

- PRV00001: 66.15% (39 claims)
- PRV98765: 66.17% (13 claims)
- PRV11223: 67.20% (5 claims)

Insights: Provider reimbursement efficiency varies widely (~66% to ~92%), indicating differences in contracts, billing behaviour, and adjustment rates.



Business Implications

This analysis enables healthcare insurers to:

- Identify **high-risk members** for early intervention
- Focus cost controls on **inpatient utilization**
- Monitor **high-cost CPT procedures** for utilization review
- Support **provider contract optimization**
- Improve reimbursement efficiency tracking

Assumptions and Limitations

1.The dataset is synthetic and does not represent real patient data

2.Clinical severity and appropriateness are not evaluated

3.CPT and provider analyses exclude rare procedures using a ≥5-claim threshold

Conclusion:

This project demonstrates an end-to-end healthcare analytics workflow:

Data cleaning → validation → visualization → actionable insights

The findings highlight that healthcare costs are driven by severity, procedure intensity, and a small subset of members, rather than claim volume alone.